Signed future terminal trace excess, robust graph complement, and tail-only headroom boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

R-138 proved two exact but apparently competing facts. Its complete future current–trace block telescopes as a signed last-insertion owner, while moving an insertion-anchored positive FAR estimate back to the reveal weight costs $2^{2\gamma(k-r)}$ and leaves a reveal-FAR/insertion-near wedge. The present result resolves the choice of coordinates before attempting another positive estimate.

The correct future operation is to sum *all* future insertion endpoints while the reveal root and output shell remain fixed. The insertion label then disappears exactly. The reveal-FAR future block becomes one weighted terminal-minus-prefix trace-excess secant, so the R-138 causal reanchoring factor is absent. This does not retract the R-138 no-go: the factor remains mandatory for a strategy which first separates nonnegative insertion-FAR cells.

The endpoint reduction also exposes two further obligations. First, the centered Doob current is not the complete R-123 packet: its predictable mean, future variance, and R-063 forest coordinate must be restored exactly once. Second, tail decay cannot create the strict balanced/low headroom required by R-129. The present note gives the exact robust shifted-Douglas complement criterion, first in the ambient finite-dimensional space and then on the actual production tangent graph.

The result is fixed finite cutoff, fixed positive floor, and fixed legal temporally faithful chart. Same-root visits and equal-time children are recombined before the formulas are used. Output projections are deterministic, mutually orthogonal, contraction-closed, and exhaustive unless an explicit R-125 complement leakage is retained. Heat satisfies the R-104 legal disintegration hypothesis. All physical and source spaces in the operator theorems are finite-dimensional; the bounded-operator statements therefore have no hidden closed-range issue.

No cutoff-, floor-, chart-, control-, filtration-, revisit-, cluster-, or refinement-uniform production trace-excess estimate is proved. No positive production graph margin, complete low/matching theorem, absolute anchor, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson estimate, removal, interacting measure, Sector-A closure, or tier promotion follows.

2. Complete root/output endpoint coordinate

Let $e = 1, \dots, N$ be the ordered control insertion events on a fixed chart, and let $J^{(e)}$ and $B^{(e)} = C_6(W^{(e)})^* C_6(W^{(e)})$ be the corresponding fully recombined current and trace coefficient. Put

$$P_r = \mathbb{E}[\,\cdot \mid \mathcal{F}_r], \qquad d_r = P_r - P_{r-1}, \tag{2.1}$$

and let $n(r)$ be the last insertion event at or before reveal r. Let $\{\Pi_m\}$ be the declared orthogonal physical-output resolution. Since Π_m is deterministic, it commutes with P_r and d_r.

For endpoint $q \in \{0, \dots, N\}$ define

$$\Phi^q_{r,m} = \Pi_m P_r J^{(q)}, \qquad b^q_{r,m} = \Pi_m P_{r-1} J^{(q)}, \tag{2.2}$$

$$Y^q_{r,m} = \Phi^q_{r,m} - b^q_{r,m} = \Pi_m d_r J^{(q)}. \tag{2.3}$$

The once-owned projected primitive trace is

$$\Theta^q_{r,m} = \mathbb{E}_{f_r, G'_r}\left[\left\|\Pi_m C_6(W^{(q)})\partial G'_r\right\|^2 \mid \mathcal{F}_r\right], \tag{2.4}$$

where f_r is the legal future beyond reveal r and G'_r has exactly the owner covariance $\Delta\Gamma_r$. Equivalently, the legal future average is P_r applied to the fresh-trace expectation. This conditioning is load-bearing for the R-125 bridge below. There is no fresh trace packet at a same-root subvisit. Define the centered energy coordinate

$$\boxed{K^q_{r,m} := \frac{1}{2}\mathbb{E}\left(\|Y^q_{r,m}\|^2 - \Theta^q_{r,m}\right).} \tag{2.5}$$

The spatialized future insertion owner is

$$\begin{aligned}\omega_{e,r,m} := \mathbb{E}\,\mathrm{Re}\langle Y^{e-1}_{r,m}, Y^e_{r,m} - Y^{e-1}_{r,m}\rangle + \frac{1}{2}\mathbb{E}\|Y^e_{r,m} - Y^{e-1}_{r,m}\|^2 \\ - \frac{1}{2}\mathbb{E}(\Theta^e_{r,m} - \Theta^{e-1}_{r,m}), \qquad r < t_e.\end{aligned} \tag{2.6}$$

This is R-138 equation (6.3) after one deterministic output projection. The current cross, current square, and trace increment remain one signed owner.

3. Signed future terminal-minus-prefix theorem

Theorem 3.1 (endpoint-first future telescope)

Under the hypotheses of Section 2,

$$\boxed{\omega_{e,r,m} = K^e_{r,m} - K^{e-1}_{r,m}.} \tag{3.1}$$

Consequently

$$\boxed{\sum_{e>n(r)} \omega_{e,r,m} = K^N_{r,m} - K^{n(r)}_{r,m}.} \tag{3.2}$$

For a collar $C \geq 5$ and reveal weight

$$w^{(\gamma,C)}_{r,m} = 2^{2\gamma(m-r-C)}\mathbf{1}_{\{m \geq r+C\}}, \tag{3.3}$$

the complete weighted future branch is therefore

$$\boxed{\mathfrak{I}^{(\gamma,C)}_{\mathrm{fut}} = \sum_{r>\ell}\sum_{m\geq r+C} w^{(\gamma,C)}_{r,m}\left(K^N_{r,m} - K^{n(r)}_{r,m}\right).} \tag{3.4}$$

No physical insertion label k occurs in (3.4), and hence no factor $2^{2\gamma(k-r)}$ occurs.

Proof. For $x = Y^{e-1}_{r,m}$ and $y = Y^{e}_{r,m} - Y^{e-1}_{r,m}$, polarization gives

$$\mathrm{Re}\langle x, y\rangle + \frac{1}{2}\|y\|^2 = \frac{1}{2}(\|x+y\|^2 - \|x\|^2). \tag{3.5}$$

Subtract the matching primitive-trace increment. This is exactly (3.1). The finite sum over every $e > n(r)$ cancels each intermediate endpoint and proves (3.2). The weight (3.3) depends only on the fixed pair (r, m), so it may multiply the already telescoped identity. Summing in (r, m) proves (3.4). ☐

Equation (3.4) is invariant under a representation-preserving subdivision which preserves the filtration, reveal order, admissible terminal and prefix controls, and output resolution. The individual $\omega_{e,r,m}$ need not be invariant. A new root, a changed reveal order, or an enlarged admissible source space is a different chart and is not covered by this assertion.

4. Exact weighted trace-excess target

Since $Y^q_{r,m} = \Pi_m d_r J^{(q)}$ is a Doob increment, its strict-past mean vanishes. Define the R-123 centered trace excess

$$D^q_{0;r,m} = \mathbb{E}\left[\Theta^q_{r,m} - \|Y^q_{r,m}\|^2 \mid \mathcal{F}_{r-1}\right]. \tag{4.1}$$

Then

$$\boxed{K^q_{r,m} = -\frac{1}{2}\mathbb{E}D^q_{0;r,m}.} \tag{4.2}$$

Combining (3.4) and (4.2) gives

$$\boxed{\mathfrak{I}^{(\gamma,C)}_{\mathrm{fut}} = -\frac{1}{2}\sum_{r>\ell}\sum_{m\geq r+C} w^{(\gamma,C)}_{r,m}\mathbb{E}(D^N_{0;r,m} - D^{n(r)}_{0;r,m}).} \tag{4.3}$$

Thus the desired lower bound

$$\mathfrak{I}^{(\gamma,C)}_{\mathrm{fut}} \geq -\eta X_{\mathrm{src}} - \zeta Y_6 - C_0 \tag{4.4}$$

is exactly equivalent to

$$\boxed{\sum_{r>\ell}\sum_{m\geq r+C} w^{(\gamma,C)}_{r,m}\mathbb{E}(D^N_{0;r,m} - D^{n(r)}_{0;r,m}) \leq 2\eta X_{\mathrm{src}} + 2\zeta Y_6 + 2C_0.} \tag{4.5}$$

No absolute value and no eventwise positive square appears in (4.5).

Theorem 3.1 is an exact reduction, not a proof of (4.5). Existing R-102 regular no-revisit estimates do not contain the weighted complete trace and forest. R-123 proves that the full production target is a trace-excess bound, but it does not establish this endpoint secant uniformly.

5. Mean, future variance, forest, and low ownership

The complete R-123 packet uses $\Phi = b + Y$, not the centered Y alone. The R-125 bridge, in the no-outer-half forest convention, is

$$\mathfrak{F}^{\mathrm{ad}}_{063,q} = \|\Phi_q\|^2 - \Theta_q + \mathrm{Var}_f\, X_q. \tag{5.1}$$

Orthogonality of b_q and Y_q gives

$$\boxed{\mathcal{P}_{\mathrm{comp},q} = \frac{1}{2}\mathbb{E}(\mathfrak{F}^{\mathrm{ad}}_{063,q} - \mathrm{Var}_f\, X_q),} \tag{5.2}$$

$$\boxed{K_q = \frac{1}{2}\mathbb{E}(\mathfrak{F}^{\mathrm{ad}}_{063,q} - \mathrm{Var}_f\, X_q - \|b_q\|^2).} \tag{5.3}$$

Equivalently,

$$\boxed{\mathbb{E}D_{0,q} = \mathbb{E}(\mathrm{Var}_f\, X_q + \|b_q\|^2 - \mathfrak{F}^{\mathrm{ad}}_{063,q}).} \tag{5.4}$$

The corresponding conditional statement places $\mathbb{E}[\,\cdot \mid \mathcal{F}_{r-1}]$ around the entire right-hand side; (5.4) is the scalar expected coordinate used in the weighted theorem. Therefore a future estimate for K alone is not yet an estimate for the complete packet. The same weighted endpoint difference of the predictable mean must be returned from its low/injected owner:

$$\boxed{\mathfrak{I}^{(\gamma,C)}_{\mathrm{fut}} + \frac{1}{2}\sum_{r,m} w^{(\gamma,C)}_{r,m}\mathbb{E}\Delta\|b_{r,m}\|^2 = -\frac{1}{2}\sum_{r,m} w^{(\gamma,C)}_{r,m}\mathbb{E}\Delta(D_0 - \|b\|^2).} \tag{5.5}$$

By (5.1), the right side is also one half of the weighted endpoint difference of $\mathfrak{F}^{\mathrm{ad}}_{063} - \mathrm{Var}_f\, X$. These are alternate coordinates of one owner. The forest is never an additional reserve.

The mean is load-bearing. In the bounded R-123 six-row fixture, with $s = 339/(8000P)$, $t = 2$, and $A(\xi) = a + \beta\sin(2\xi)$,

$$\|b\|^2 = 16s\beta^2 e^{-4}, \tag{5.6}$$

$$D_0 = 16s\beta^2(e^{-4} - 2e^{-8}) > 0, \tag{5.7}$$

$$\mathbb{E}\mathcal{P}_{\mathrm{comp}} = 16s\beta^2 e^{-8} > 0, \tag{5.8}$$

but

$$\boxed{K = -\frac{1}{2}D_0 < 0.} \tag{5.9}$$

Thus positivity of the complete packet does not imply positivity of its future-centered coordinate. This is a coordinate counterfixture, not a counterexample to (4.5) for the production aggregate.

The still-missing production theorem is an intertwiner matching the weighted b-difference in (5.5) with exactly one R-079 low/injected owner, without a root, visit, output, or refinement multiplicity.

6. Why a wedge-only telescope is impossible

Let k_e be the physical grade of insertion e and mask only the reveal-FAR/insertion-near wedge by

$$\chi_e(r,m) = \mathbf{1}_{\{r+C\le m<k_e+C\}}. \tag{6.1}$$

For any endpoint sequence K_e finite summation by parts gives

$$\begin{aligned}\sum_{e=a}^{N}\chi_e(K_e - K_{e-1}) = \chi_N K_N - \chi_a K_{a-1}\\ + \sum_{e=a}^{N-1}(\chi_e - \chi_{e+1})K_e.\end{aligned} \tag{6.2}$$

The last sum vanishes for the complete mask $\chi_e \equiv 1$, but not for a moving wedge. If K_e alternates $0, 1, 0, 1, \ldots, 0$ and the mask selects only rising increments, the terminal difference is zero while the masked sum grows linearly with the number of rises. This establishes `NG-2026-07-31-A13-WEDGE-ONLY-FUTURE-TELESCOPE`. It is a finite algebraic method no-go, not a production-action counterexample.

The same obstruction appears as a cross-owner. With trace zero and $y_F = u$, $y_N = -u$,

$$\frac{1}{2}\|y_F + y_N\|^2 = 0, \tag{6.3}$$

while the positive far square is $\|u\|^2/2$ and the cross-retaining near owner is $-\|u\|^2/2$. Spending the far square separately consumes a cancellation which the near owner still needs. A legal positive split would require a new Bessel, Loewner, or causal Schur theorem for the complete production columns.

7. Physical Riesz response: reused theorem and projector firewall

R-128–R-131 already contain the canonical physical response. It must not be registered again. The notation needs one firewall. Let P_{pred} be the orthogonal predictable source projection and define the actual synthesis

$$M_\pi = L_\pi P_{\mathrm{pred}} : \bar{\mathcal{S}}_\pi \longrightarrow \mathcal{X}. \tag{7.1}$$

If the fully recombined physical Hessian has bounded selfadjoint Riesz representative B_h, put

$$A_\pi = B_h M_\pi, \qquad H_\pi = M_\pi^* B_h M_\pi. \tag{7.2}$$

Then

$$\boxed{M_\pi^* A_\pi = A_\pi^* M_\pi = H_\pi = H_\pi^*.} \tag{7.3}$$

In ambient notation the legal identities are

$$\boxed{P_{\mathrm{pred}} L_\pi^* A_\pi = A_\pi^* L_\pi P_{\mathrm{pred}} = H_\pi.} \tag{7.4}$$

The shorter formula without P_{pred} is valid only when L_π already denotes the restricted predictable map.

For orthogonal shell analysis $\mathcal{C}z = (\Pi_m z)_m$, set $R = \mathcal{C}M_\pi$ and $F = \mathcal{C}A_\pi$. R-129/R-131 give

$$R^* F = F^* R = H_\pi. \tag{7.5}$$

Let $\{E_b\}$ be an exhaustive orthogonal resolution of the predictable source space, so $E_b E_{b'} = \delta_{bb'} E_b$ and $\sum_b E_b = \mathrm{I}$. Define

$$T_{mb} = M_\pi^* \Pi_m A_\pi E_b, \qquad K_{mb} = \frac{1}{2}(T_{mb} + T_{mb}^*). \tag{7.6}$$

Then

$$\boxed{H_\pi = \sum_{m,b} K_{mb}.} \tag{7.7}$$

This is the nonduplicating forward/legal-reverse ledger. The true adjoint has coefficient 1/2 and is not a second geometric owner.

Finite spatial cutoff alone does not prove that $B_h M_\pi$ extends boundedly in the production norm. For example, multiplication by ξ^2 is finite on every bounded Gaussian cylinder but unbounded on $L^2(\gamma)$: normalized indicators of $\{|\xi| > n\}$ have quadratic form at least n^2. The missing analytic hypothesis is the cutoff/chart/refinement-uniform bounded forward response, not the algebraic reverse.

8. Exact robust shifted-Douglas complement

Let X, Y, L be finite-dimensional real Hilbert spaces. Let

$$\mathcal{M}(T) = \begin{pmatrix} E & -(A_0+T)/2 & -B \\ -(A_0+T)^*/2 & F & -C \\ -B^* & -C^* & D \end{pmatrix}, \tag{8.1}$$

where $T : Y \to X$ is the tail response and A_0 contains every balanced, near, and future-wedge cross which is not in that tail. Put

$$K = \begin{pmatrix} B \\ C \end{pmatrix}, \qquad M_{2,0} = \begin{pmatrix} E & -A_0/2 \\ -A_0^*/2 & F \end{pmatrix}. \tag{8.2}$$

Fix a desired gap $\mu \geq 0$, set $D_\mu = D - \mu\mathrm{I}$, and assume

$$D_\mu \succeq 0, \qquad \operatorname{Ran} K^* \subseteq \operatorname{Ran} D_\mu^{1/2}. \tag{8.3}$$

Let $D_\mu^{1/2} X_\mu = K^*$ be the Douglas reduced solution and define

$$S_\mu = M_{2,0} - \mu\mathrm{I} - X_\mu^* X_\mu. \tag{8.4}$$

Theorem 8.1 (robust complement criterion)

For $\tau \geq 0$, the following are equivalent:

$$\mathcal{M}(T) \succeq \mu \mathrm{I} \quad \text{for every } T : Y \to X \text{ with } \|T\| \leq \tau, \tag{8.5}$$

and

$$\boxed{\langle (x,y), S_\mu(x,y) \rangle \geq \tau \|x\| \|y\| \quad \text{for every } x \in X,\ y \in Y.} \tag{8.6}$$

Proof. Shift (8.1) by μI and apply the finite-dimensional Douglas/Schur criterion to the low block. The remaining tail contribution to the reduced quadratic form is $-\operatorname{Re}\langle x, Ty\rangle$. For every $\|T\| \leq \tau$,

$$|\operatorname{Re}\langle x, Ty\rangle| \leq \tau \|x\| \|y\|, \tag{8.7}$$

so (8.6) is sufficient. Conversely, for fixed nonzero x, y, the rank-one map sending $y/\|y\|$ to $\tau x/\|x\|$ attains the right side of (8.7). Thus robustness for every such T implies (8.6). ☐

For one known tail T_C, the still weaker exact condition is simply the positivity of (8.4) after subtracting its symmetrized tail block. The scalar R-133/R-138 headroom follows as a conservative corollary. If

$$E \succeq e\mathrm{I}, \quad F \succeq f\mathrm{I}, \quad D \succeq d\mathrm{I}, \quad \|K\| \leq k, \quad \|A_0\| \leq a_0, \tag{8.8}$$

then, for the zero-gap case $\mu = 0$, with $\sigma = k^2/d$, the condition

$$\boxed{a_0 + \tau < 2\sqrt{(e-\sigma)(f-\sigma)}} \tag{8.9}$$

is sufficient when $e, f > \sigma$. It is not the weakest condition because it replaces the reduced low operator by the scalar bound σI. For a prescribed gap $\mu > 0$, the corresponding sufficient conditions are

$$d > \mu, \qquad \sigma_\mu = \frac{k^2}{d-\mu}, \qquad e - \mu > \sigma_\mu, \quad f - \mu > \sigma_\mu, \tag{8.10}$$

and

$$\boxed{a_0 + \tau < 2\sqrt{(e-\mu-\sigma_\mu)(f-\mu-\sigma_\mu)}.} \tag{8.11}$$

9. Tail-only and low-only strict-gap no-go

No estimate $\|T_C\| \leq \tau_C$ with $\tau_C \downarrow 0$ can by itself imply a strict gap. Take $X = Y = L = \mathbb{R}$, $B = C = 0$, $D = d > 0$, and

$$E = e > 0, \qquad F = f > 0, \qquad A_0 = 2\sqrt{ef}, \qquad T_C = 0. \tag{9.1}$$

Every tail estimate holds with charge zero, but

$$\begin{pmatrix} e & -\sqrt{ef} \\ -\sqrt{ef} & f \end{pmatrix} \begin{pmatrix} \sqrt{f} \\ \sqrt{e} \end{pmatrix} = 0. \tag{9.2}$$

Choosing $A_0 > 2\sqrt{ef}$ even destroys semidefiniteness. This establishes `NG-2026-07-31-A13-TAIL-ONLY-SHIFTED-DOUGLAS-HEADROOM`. It is a logical non-identifiability theorem, not a production counterexample.

Low transversality is independent. With

$$E = F = D = 1, \qquad A_0 = T_C = C = 0, \qquad B = 1, \tag{9.3}$$

the full matrix has kernel vector $(1, 0, 1)$. Hence a positive balanced margin cannot replace the complete low range and Schur conditions.

10. Exact production-graph version

The ambient criterion may be stronger than needed. Let the actual source-to-channel tangent map be

$$G_{t,h}u = (X_{t,h}u, Y_{t,h}u, L_{t,h}u). \tag{10.1}$$

Let $\mathcal{M}_0(C)$ contain every once-owned non-tail, balanced, wedge, diagonal, and low block. The tail enters only through $-\operatorname{Re}\langle X_{t,h}u, T_C Y_{t,h}u\rangle$.

Theorem 10.1 (graph-robust complement)

If $\|T_C\| \le \tau_C$, then

$$\boxed{\langle G_{t,h}u, \mathcal{M}_0(C)G_{t,h}u\rangle \ge \tau_C\|X_{t,h}u\|\|Y_{t,h}u\| + \mu\|u\|^2} \tag{10.2}$$

for every legal chart, Taylor point t, base control h, and source direction u implies

$$G^*_{t,h}\mathcal{M}(T_C)G_{t,h} \succeq \mu\mathrm{I}. \tag{10.3}$$

If robustness is required for every tail operator of norm at most τ_C, (10.2) is also necessary.

Proof. The tail quadratic form has absolute value at most the product on the right of (10.2), proving sufficiency. For necessity at a fixed u, choose the rank-one tail which attains that product, exactly as in Theorem 8.1. ☐

The graph theorem is the smallest operator target presently visible. It does not demand positivity in directions which no legal production source can reach. It nevertheless keeps all complete owners and requires one uniform $\mu > 0$. The affine baseline $P(0) + DP(0)[h]$ and the absolute low anchor are not consequences of a Hessian graph inequality and remain separate payments.

11. Smallest non-circular successor

The active successor remains `A13-CLASSII-COMPLETE-RAW-DOOB-DIRECT-FUTURE-SPATIAL-ONE-USE-BOUND`; R-139 narrows its future and complement coordinates rather than creating a new claim card. The R-138 future fork may now be shortened. At one fixed collar $C_0 \ge 5$:

1. prove the complete weighted endpoint trace-excess secant

$$\sum_{r,m} w^{(\gamma,C_0)}_{r,m}\mathbb{E}\Delta(D_0 - \|b\|^2) \le 2\eta X_{\mathrm{src}} + 2\zeta Y_6 + 2C_0', \tag{11.1}$$

   with the mean/low owner, future variance, and forest each used once;

2. identify its physical response with the R-131 canonical response and prove one tail norm $\|T_{\mathrm{tail}}(C_0)\| \le \tau_{C_0}$;

3. prove the single production-graph complement inequality (10.2), rather than separately guessing $C_{\mathrm{mix}}, C_{\mathrm{far}}, c_{\mathrm{bal}}, R_0, S_0, d, k$;

4. spend the remaining strict margin on the finite-low affine baseline and absolute anchor exactly once.

Equation (11.1) is the analytic burden. Theorem 3.1 removes only an unnecessary causal-depth tax; it does not supply the weighted secant bound. Theorems 8.1 and 10.1 identify the exact acceptance test; they do not supply positive production headroom.

The entropy/Boue–Dupuis restatement is not a successor. The R-087/R-093 CORE equality already identifies the uniform source-union lower bound with the open $q = 10/9$ Nelson moment. Replacing (11.1) by that moment would be circular.

12. Proof and failure map

Route	Verdict	Reusable endpoint or boundary
Complete insertion sum at fixed (r, m)	proved exactly	$K^N_{r,m} - K^{n(r)}_{r,m}$; no $k - r$ factor
Weighted future trace excess	reduced exactly	Equation (4.5); production estimate open
Centered future D_0 alone	rejected	Restore weighted b, variance, forest, and low owner
Wedge-only endpoint telescope	failed	Moving mask leaves internal variation (6.2)
Positive far square as reserve	failed	Cross-retaining near owner cancels it
Canonical physical Riesz response	already R-131	Use restricted predictable synthesis $M = LP_{\text{pred}}$
Forward plus legal reverse	exact one owner	$\frac{1}{2}(T + T^*)$, not two payments
Tail decay creates headroom	failed	Balanced and low zero-gap fixtures preserve every tail bound
Ambient robust complement	proved exactly	Theorem 8.1
Production-graph complement	proved exactly as criterion	Theorem 10.1; uniform positive margin open
Entropy restart	circular	Already the open $q = 10/9$ Nelson objective

13. Devil's-advocate review

1. **Objection: R-139 disproves the R-138 reanchoring factor. DISMISSED.** The factor is exact for a separated insertion-FAR positive ledger. R-139 avoids that ledger by summing the complete signed future endpoint first.

2. **Objection: the future branch is now proved. UPHELD AGAINST THE CLAIM.** The exact remaining estimate is (4.5), or the complete-owner form (11.1). No authority presently proves it uniformly.

3. **Objection: D_0 is the full R-123 packet. UPHELD AGAINST THE CLAIM.** Equations (5.2)–(5.5) and the R-123 fixture show that the predictable mean is load-bearing.

4. **Objection: the R-063 forest may be added to improve the sign. UPHELD AGAINST THE CLAIM.** It is the alternative coordinate in (5.1), not a new reserve.

5. **Objection: the wedge can be telescoped separately. UPHELD AGAINST THE CLAIM.** Equation (6.2) leaves the mask-variation sum; the alternating endpoint fixture makes it grow.

6. **Objection: R-139 newly constructs the physical Riesz response. DISMISSED.** R-128–R-131 already do so conditionally. Section 7 only records the predictable-projection notation and the still-open bounded production extension.

7. **Objection: a tail estimate eventually gives strict positivity. UPHELD AGAINST THE CLAIM.** The zero-tail balanced fixture (9.1) has zero gap at every collar.

8. **Objection: balanced positivity makes the low block harmless. UPHELD AGAINST THE CLAIM.** Fixture (9.3) has a low-induced kernel.

9. **Objection: ambient positivity is necessary. DISMISSED.** Only the source graph is physical. Theorem 10.1 states the exact robust graph criterion, but does not claim the graph inequality.

10. **Objection: the graph Hessian proves the absolute action bound. UPHELD AGAINST THE CLAIM.** The affine finite-low baseline and absolute anchor still require one separate payment.

14. Executed evidence

The primary executable checks an exact five-endpoint signed telescope, the reveal-weight factor, the $-1/2$ trace-excess sign, forest/variance/mean ownership, the R-123 sign fixture, a masked-wedge counterfixture, coherent far/near cancellation, balanced and low zero-gap fixtures, the robust scalar condition, graph compression, and the predictable-projection firewall. The independent standard-library executable uses different endpoint, forest, mask, balanced, low, and robust-complement fixtures and imports no primary code. The integrated verifier reruns both, checks authority and public surface pins, rebuilds the PDF deterministically, performs text/security checks, and renders every page through Poppler.

15. Result footer

```
Result ID: A13-CLASSII-SIGNED-FUTURE-ENDPOINT-GRAPH-
  COMPLEMENT-BOUNDARY.
Ledger ID: R-139.
Precise statement: Theorems 3.1, 8.1, and 10.1; exact trace-excess and
  owner identities (4.3)-(5.5); masked-wedge and tail-only boundaries.
Scope: fixed finite cutoff and positive floor; fixed temporally faithful
  chart; recombined visits/equal-time children; exhaustive orthogonal
  deterministic output resolution; legal R-104 heat; finite-dimensional
  operator spaces; bounded canonical physical response only where stated.
Dependencies: A1; R-063, R-079, R-087, R-093, R-102, R-104, R-123,
  R-125, R-128, R-129, R-131, R-133, R-135, R-136, R-137, R-138.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_signed_future_endpoint_graph_
  complement_boundary_verify.py
Expected output: R-139 integrated PASS; primary, independent, note/PDF,
  manifest, exploration, negative, authority, and public-surface gates pass.
Falsification gate: any failed endpoint telescope, sign or factor-one-half,
  mean/variance/forest incidence, mask variation, coherent cross, robust
  complement equivalence, predictable projection, graph restriction,
  authority/hash, PDF, or public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no production weighted trace-excess secant,
  complete mean/low intertwiner, bounded uniform response, tail norm,
  positive graph complement margin, low/matching theorem, absolute anchor,
  A13 gate, OVERLAP_src, Nelson estimate, removal, interacting measure,
  Sector-A closure, or T5-T7 claim follows.
Next required action: at one fixed collar prove the complete-owner weighted
  Delta(D0-||b||^2) production estimate, its canonical tail response, and
  the single strict graph-complement inequality; then pay the affine low
  baseline and absolute anchor once before invoking R-129/R-093.
```